
Graphic Model: Hospital Readmission Analysis Framework

Objective

Visualize the relationship between patient, clinical, operational, and systemic factors contributing to hospital readmissions, and outline pathways for improvement.

Model Structure

1. Input Layer: Data Sources

- Patient demographics (age, gender, comorbidities)
- Admission details (type, length of stay, department)
- Clinical data (diagnoses, lab results, medications)
- Social and environmental data (support systems, socioeconomic status)

Visual Representation:

Rectangular boxes feeding into the next layer, each labeled with a data source category.

2. Process Layer: Analytical Components

- **Data Cleaning & Preparation:** Handle missing values, normalize data.
- **Exploratory Data Analysis (EDA):** Identify trends and correlations.
- **Predictive Modeling:** Use logistic regression or decision trees to predict readmission risk.
- **Visualization:** Create dashboards and charts for insights.

Visual Representation:

Flow arrows connecting from data inputs to analytical processes, forming a pipeline.

3. Output Layer: Insights & Findings

- High-risk patient groups identified
- Key diagnoses linked to readmissions
- Operational inefficiencies detected
- Medication-related risk factors highlighted

Visual Representation:

Circular or hexagonal nodes summarizing insights, branching from the process layer.

4. Outcome Layer: Recommendations & Actions

- Enhanced discharge planning
- Chronic disease management programs
- Predictive readmission monitoring system
- Staff training and communication improvement

Visual Representation:

Arrows leading from insights to actionable strategies, displayed as labeled boxes or icons (e.g., stethoscope for clinical care, chart for analytics).

5. Feedback Loop

- Continuous monitoring of readmission rates
- Integration of new data for model refinement
- Regular performance evaluation

Visual Representation:

Curved arrow looping back from outcomes to data sources, symbolizing continuous improvement.

Design Notes for Miro or Canva

- Use **layered horizontal flow** (left to right) to show progression from data to action.
 - Apply **color coding**:
 - Blue – Data Inputs
 - Green – Analysis
 - Orange – Insights
 - Purple – Recommendations
 - Gray – Feedback Loop
 - Include icons for clarity (database, magnifying glass, chart, gear, arrow).
-

Purpose

This graphic model provides a visual framework for understanding how data-driven analysis leads to actionable strategies that reduce hospital readmissions and improve patient outcomes.